A2 · Unit 4 · WCH14 Rates, Equilibria and Further Organic Chemistry

Acid-base equilibria

Acids and bases are all about protons moving from one particle to another. This topic turns that idea into numbers: the pH scale, the ionic product of water, the acid dissociation constant K_a, the shapes of titration curves, and the buffers that keep the pH of your blood almost perfectly steady.

SPEC 14.1 Brønsted–Lowry acids and bases

WHAT THIS SPEC POINT ASKS

Know that a Brønsted–Lowry acid donates a proton, a base accepts one, and that an acid-base reaction is a proton transfer.

The definitions

- A **Brønsted–Lowry acid** is a **proton donor**. A **Brønsted–Lowry base** is a **proton acceptor**.
- "Proton" here means an **H^+ ion**: a hydrogen atom that has lost its electron. Never write "hydrogen atom".
- An acid-base reaction is a **proton transfer** from the acid to the base. You can't have one without the other: the acid only donates if something accepts.

Examples

Reaction	Acid (donates H^+)	Base (accepts H^+)
$HCl(g) + H_2O(l) \rightarrow H_3O^+(aq) + Cl^-(aq)$	HCl	H_2O
$NH_3(aq) + H_2O(l) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$	H_2O	NH_3
$HCl(g) + NH_3(g) \rightarrow NH_4Cl(s)$	HCl	NH_3
$CH_3COOH(aq) + OH^-(aq) \rightarrow CH_3COO^-(aq) + H_2O(l)$	CH_3COOH	OH^-

- In water, acids form the **oxonium ion H_3O^+** (also called hydronium or hydroxonium). $H^+(aq)$ is shorthand for $H_3O^+(aq)$; mark schemes accept either.
- **Water can act as an acid or a base**, depending on its partner: it accepts a proton from HCl but donates one to NH_3. Species like this are **amphoteric** (other examples: HCO_3^-, $H_2PO_4^-$, HSO_4^-).

MARK-SCHEME WORDING

- "proton donor" / "proton acceptor"; "transfer of a proton / H^+".
- An acid "donates hydrogen ions and becomes an anion" (for a neutral acid like HCl or CH_3COOH).
- In an MCQ, check that the species you pick acts as an acid (or base) on the side of the equation the question means.

Often asked as: MCQ "which species is acting as a Brønsted–Lowry base?", identify the acid and base in a given equation, role of water.

COMMON MISTAKE

- Writing "hydrogen atom" or "hydrogen" for proton.
- Thinking water is always neutral in these reactions: in $NH_3 + H_2O$, water is the **acid**.
- Putting a charge on only one product of a proton transfer when both change charge ($HCl + H_2O$ gives H_3O^+ **and** Cl^-).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- An acid-base **indicator** is itself a weak acid, $HIn \rightleftharpoons H^+ + In^-$, with different colours for HIn and In^- (WCH14 Oct 2024).
- A carboxylic acid can act as a **base** when mixed with a stronger acid: it accepts a proton to become $RCOOH_2^+$ (WCH14 Jun 2025). See 14.2.

SPEC 14.2 Conjugate acid-base pairs

WHAT THIS SPEC POINT ASKS

Identify the conjugate acid-base pairs in a Brønsted–Lowry reaction.

What a conjugate pair is

- When an acid loses a proton, what's left can accept one back: it is the **conjugate base** of that acid. When a base gains a proton, it becomes its **conjugate acid**.
- The two members of a pair **differ by exactly one H^+**. Every acid-base equilibrium contains **two** pairs:

acid 1 + base 2 $\rightleftharpoons$ base 1 + acid 2

Worked pairs

Equation	Pair 1 (acid / conjugate base)	Pair 2 (conjugate acid / base)
$CH_3COOH + H_2O \rightleftharpoons CH_3COO^- + H_3O^+$	CH_3COOH / CH_3COO^-	H_3O^+ / H_2O
$NH_3 + H_2O \rightleftharpoons NH_4^+ + OH^-$	H_2O / OH^-	NH_4^+ / NH_3
$HNO_3 + H_2SO_4 \rightleftharpoons H_2NO_3^+ + HSO_4^-$	H_2SO_4 / HSO_4^-	$H_2NO_3^+$ / HNO_3
$H_2PO_4^- + OH^- \rightleftharpoons HPO_4^{2-} + H_2O$	$H_2PO_4^-$ / HPO_4^{2-}	H_2O / OH^-

- In the third row, nitric acid is acting as a **base**, because sulfuric acid is the stronger acid. The **stronger acid** is always the one that donates.

How to label pairs in an answer

1. Find the two species on each side that differ by one H^+.
2. Label them **acid 1 / base 1** and **acid 2 / base 2**, or join each pair with a line. Lines or the words "conjugate acid" and "conjugate base" are safest.
3. Check the charges: removing H^+ lowers the charge by one (neutral acid → 1− base; 1+ acid → neutral base).

WORKED EXAMPLE

Worked example: one weak acid protonating another

Methanoic acid (pKa 3.75) is mixed with propanoic acid (pKa 4.87) in an unreactive solvent. Write the equation and label the pairs.

The **smaller pKa** means methanoic acid is the **stronger acid**, so it donates and propanoic acid accepts:

$HCOOH + CH_3CH_2COOH \rightleftharpoons HCOO^- + CH_3CH_2COOH_2^+$

acid 1 + base 2 ⇌ base 1 + acid 2

Pairs: $HCOOH$ / $HCOO^-$ and $CH_3CH_2COOH_2^+$ / CH_3CH_2COOH.

MARK-SCHEME WORDING

- Pairs credited when "labelled B1 A2 A1 B2 or joined by lines"; any clear indication of linkage.
- The equation must have correct charges: $CH_3COOH + CH_2ClCOOH \rightleftharpoons CH_3COOH_2^+ + CH_2ClCOO^-$ (chloroethanoic acid is stronger, so it donates).
- Each correct pair is often one mark; all four species identified correctly is the second.

Often asked as: identify both pairs in an equation; write the equation for one acid reacting with another (use the Ka/pKa table to decide which donates); MCQ on the conjugate base of a given species.

COMMON MISTAKE

- Linking an acid with the **other** acid instead of with its own conjugate base.
- Getting charges wrong: the conjugate base of a neutral acid is negative, and the conjugate base of a positive acid (like pyridinium, $C_5H_5NH^+$) is **neutral**.
- Not using the Ka table that the question gives: it tells you which acid is stronger and so which one donates.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Carboxylic acids as bases**: protonated acids such as $CH_3COOH_2^+$ appear in equations when two acids are mixed, and this has come up on four papers (WCH14 Jun 2024).
- Conjugate pairs in **phosphate** systems: $H_2PO_4^-$ / HPO_4^{2-} (WCH14 Oct 2020).

SPEC 14.3 Definition of pH

WHAT THIS SPEC POINT ASKS

Define pH.

The definition

$$\mathbf{pH = -\log_{10}[H^+(aq)]}$$

- In words: **pH is the negative logarithm, to base 10, of the hydrogen ion concentration** (in mol dm^{-3}). $[H_3O^+]$ in place of $[H^+]$ is equally correct.
- Square brackets mean "concentration of … in mol dm^{-3}".
- Because it is a **log scale**, a change of **1 pH unit = a factor of 10** in $[H^+]$. pH 3 has 10 times the $[H^+]$ of pH 4 and 100 times that of pH 5.
- Lower pH = higher $[H^+]$ = more acidic.

MARK-SCHEME WORDING

- "$pH = -\log_{10}[H_3O^+]$"; ACCEPT lg or log, and $[H^+]$; or "negative log to the base 10 of the hydrogen ion / oxonium ion concentration".

- Square brackets are needed for concentration.

Often asked as: define pH (1 mark); as the first step of nearly every Topic 14 calculation.

COMMON MISTAKE

- Missing the minus sign, or using ln (natural log) instead of $\log_{10}$.
- Writing "the concentration of hydrogen" (it's hydrogen **ions**).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Negative pH**: if $[H^+]$ is greater than 1 mol dm^{-3}, the pH is negative. 2.0 mol dm^{-3} HNO_3 has pH = −log 2.0 = **−0.30** (WCH14 Oct 2025). There is nothing special about pH 0: the 0–14 scale is just the common range.

SPEC 14.4 Calculating pH from $[H^+]$

WHAT THIS SPEC POINT ASKS

Calculate pH from the hydrogen ion concentration.

Method

- Press **(−) log ($[H^+]$)** on the calculator. Enter powers of ten with the $\times 10^x$ / EXP key, not by typing "× 10 ^".
- Give pH to **2 decimal places** unless told otherwise (mark schemes accept 2 or 3 significant figures; 1 s.f. is never enough).

WORKED EXAMPLE

Worked example: pH from $[H^+]$

(a) $[H^+] = 3.2 \times 10^{-3}$ mol dm^{-3}. pH = $-\log(3.2 \times 10^{-3})$ = **2.49**.

(b) $[H^+] = 2.0$ mol dm^{-3}. pH = $-\log(2.0)$ = **−0.30**. A negative pH is fine.

(c) $[H^+] = 1.0 \times 10^{-9}$ mol dm^{-3}. pH = **9.00**. This solution is alkaline; it still contains some H^+.

EXAM TIP

Sense-check every answer. An acid must have pH below 7 (at 25 °C), and a stronger or more concentrated acid must have a lower pH. Mark schemes refuse any pH ≥ 7 for an acid even with error carried forward.

MARK-SCHEME WORDING

- 2–3 s.f. acceptable (e.g. "2.68 or 2.7"); IGNORE s.f. except 1 s.f.
- "TE … but do not award a pH ≥ 7" for an acid solution.

Often asked as: the last step of weak acid, buffer and mixture calculations; MCQ with distractors that forget the minus sign.

COMMON MISTAKE

- Forgetting to take the square root of Ka × [HA] **before** taking the log (weak acids, 14.9).
- Rounding $[H^+]$ early, then taking the log: keep the full calculator value.

SPEC 14.5 Calculating $[H^+]$ from pH

WHAT THIS SPEC POINT ASKS

Calculate $[H^+]$ in mol dm^{-3} from pH using $[H^+] = 10^{-pH}$.

Method

$$[H^+] = 10^{-pH}$$

- On the calculator: **SHIFT log** (10^x) then **(−) pH**.
- The answer has units of **mol dm^{-3}**.

WORKED EXAMPLE

Worked example: $[H^+]$ from pH, and diluting to a target pH

(a) pH = 3.40. $[H^+] = 10^{-3.40}$ = **4.0×10^{-4} mol dm^{-3}**.

(b) 25.0 cm^3 of HCl has pH 1.00. What volume of water must be added to raise the pH to 2.00?

$[H^+]$ goes from 0.100 to 0.0100 mol dm^{-3}: a tenfold dilution (HCl is fully dissociated, so $[H^+] \propto$ concentration).

Final volume = 25.0 × 10 = 250 cm³, so add **225 cm³** of water.

MARK-SCHEME WORDING

- Evaluate the power: "$[H^+] = 10^{-1.125} = 0.075$ mol dm^{-3}". Writing 10^−1.125 and stopping doesn't score.
- Dilution questions: final volume first, then subtract the starting volume for the water added (2–3 s.f.).

Often asked as: $[H^+]$ from a measured pH as the first step of finding Ka (14.14); dilution to a given pH.

COMMON MISTAKE

- Rounding $10^{-0.97} = 0.107$ to "0.1" and then doubling for sulfuric acid. Use the value the pH gives you.
- Giving the final volume when the question asked for the volume of water **added**.

SPEC 14.6 Strong and weak acids: degree of dissociation

WHAT THIS SPEC POINT ASKS

Understand the difference between strong and weak acids in terms of how far they dissociate.

Strong v weak

- A **strong acid** is (almost) **completely dissociated** into ions in water. Use a one-way arrow:
  $HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$
  Examples: HCl, HNO_3, HBr, HI, H_2SO_4 (first proton).
- A **weak acid** is only **partially dissociated** (slightly ionised). Most molecules stay as HA, so use ⇌:
  $CH_3COOH(aq) \rightleftharpoons CH_3COO^-(aq) + H^+(aq)$
  Examples: carboxylic acids, HCN, H_2CO_3, NH_4^+, HF. In 0.1 mol dm^{-3} ethanoic acid only about 1 molecule in 80 is dissociated.
- For the same concentration, the weak acid has a **lower $[H^+]$**, so a **higher pH**, lower electrical conductivity and slower reactions with metals or carbonates. The total amount of acid that a base can neutralise is the same.

Strength is not concentration

	Meaning
Strong / weak	**how far** the acid dissociates (a fixed property at a given temperature)

	Meaning
Concentrated / dilute	**how much** acid is dissolved per dm³

So 0.001 mol dm^{-3} HCl is a **dilute strong** acid; 10 mol dm^{-3} ethanoic acid is a **concentrated weak** acid.

Sulfuric acid

- $H_2SO_4 \rightarrow H^+ + HSO_4^-$ is **complete** (strong).
- $HSO_4^- \rightleftharpoons H^+ + SO_4^{2-}$ is **partial**, and the high [H^+] from the first step pushes this second equilibrium to the left (suppresses it).
- So 0.100 mol dm^{-3} H_2SO_4 has [H^+] a little above 0.100 mol dm^{-3} (measured pH about 0.97), not 0.200.

MARK-SCHEME WORDING

- "a weak acid is dissociated (into its ions) to a small extent and a strong acid is (almost) completely dissociated"; ALLOW "partially ionised" / "fully ionised".
- A weak acid solution "contains a significant proportion of undissociated acid".
- Sulfuric acid: "the first ionisation is complete … the high [H^+] from the first suppresses the second equilibrium".

Often asked as: define a weak acid (1–2 marks); compare the pH of equimolar strong and weak acids; explain why diluting changes their pH differently (14.13).

COMMON MISTAKE

- Describing pH or "position of equilibrium" instead of **dissociation/ionisation** when asked to define strong and weak.
- Mixing up "weak" with "dilute".
- Vague sulfuric acid answers that don't say **which** of its two equilibria is complete.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Inductive effects**: an electronegative atom near the COOH group makes the acid **stronger**. In chloroethanoic acid the Cl is **electron-withdrawing**, so it pulls electron density away from the O–H bond (weakening it) and **stabilises the COO^- ion** by spreading its charge. Ka order: CH_3COOH < $CH_2ClCOOH$ < $CHCl_2COOH$ < CCl_3COOH. Alkyl groups push electrons the other way, which is why propanoic acid (pKa 4.87) is slightly weaker than ethanoic acid (pKa 4.76). "More hydrogen bonding" is not accepted as the reason (WCH14 Jun 2025).

- The extra OH in lactic acid is electron-withdrawing, which is why lactic acid is stronger than propanoic acid. The OH itself is **not** acidic (WCH14 Oct 2022).

SPEC 14.7 pH of a strong acid

WHAT THIS SPEC POINT ASKS

Calculate the pH of a strong acid.

Method

- A strong monoprotic acid is fully dissociated, so **$[H^+]$ = concentration of the acid**. Then pH = −log$[H^+]$.
- For a **mixture** of strong acid and strong base, work in **moles**:
  1. moles H^+ and moles OH^- at the start;
  2. subtract to find which is **in excess** and by how much;
  3. divide by the **total volume** (in dm^3) to get the concentration;
  4. take −log (acid in excess) or go via Kw (alkali in excess, 14.11).

WORKED EXAMPLE

Worked example: strong acid, and a strong acid–strong base mixture

(a) 0.0250 mol dm^{-3} HCl: $[H^+]$ = 0.0250, pH = **1.60**.

(b) 25.0 cm^3 of 0.100 mol dm^{-3} HCl is mixed with 15.0 cm^3 of 0.100 mol dm^{-3} NaOH.

moles H^+ = 0.100 × 25.0/1000 = 2.50 × 10^{-3} mol

moles OH^- = 0.100 × 15.0/1000 = 1.50 × 10^{-3} mol

excess H^+ = 1.00 × 10^{-3} mol in 40.0 cm^3

$[H^+]$ = 1.00 × 10^{-3} ÷ 0.0400 = 0.0250 mol dm^{-3}, pH = **1.60**.

EXAM TIP

For sulfuric acid in a **neutralisation** (moles) calculation, 1 mol H_2SO_4 supplies **2 mol H^+**. For the pH of the acid **on its own**, read the question: it may tell you to assume both protons dissociate, or give you a measured pH.

pH of a strong acid, and back again

1. 0.0250 mol dm^{-3} nitric acid: $[H^+]$ = 0.0250 mol dm^{-3}, so pH = −log 0.0250 = **1.60**.

2. A hydrochloric acid solution has pH 2.45: $[H^+] = 10^{-2.45} = 3.55 \times 10^{-3}$ mol dm^{-3}, and that is also the concentration of the acid.

MARK-SCHEME WORDING

- Mixtures: "moles of H^+ – moles of OH^-", "÷ total volume", then pH.
- With H_2SO_4 and NaOH: moles H^+ = 2 × moles H_2SO_4.

Often asked as: MCQ on the pH after a tenfold dilution or concentration change; pH of a mixture; pH after part of an acid has reacted.

COMMON MISTAKE

- Forgetting to subtract the moles that reacted, or subtracting moles of H_2SO_4 instead of moles of H^+.
- Dividing by the volume of one solution instead of the **total** volume.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Strong acid–strong base mixtures and the dibasic nature of H_2SO_4 (WCH14 Oct 2024). Mixtures with excess acid (WCH14 Oct 2025).

SPEC 14.8 The acid dissociation constant, Ka

WHAT THIS SPEC POINT ASKS

Write the expression for Ka for a weak acid.

The expression

For a weak acid HA(aq) ⇌ H^+(aq) + A^-(aq):

$$K_a = [H^+][A^-] \div [HA] \text{ units: mol dm}^{-3}$$

- For ethanoic acid: Ka = $[H^+][CH_3COO^-] \div [CH_3COOH]$.
- **Water does not appear** (it is the solvent; its concentration is effectively constant).

- All concentrations are **equilibrium** concentrations.
- **Bigger Ka = stronger acid** (more dissociated). Ka depends only on **temperature**.

Acid	Ka / mol dm^{-3}	pKa
methanoic, HCOOH	1.8×10^{-4}	3.75
benzoic, C_6H_5COOH	6.3×10^{-5}	4.20
ethanoic, CH_3COOH	1.7×10^{-5}	4.76
propanoic, CH_3CH_2COOH	1.3×10^{-5}	4.87
carbonic, H_2CO_3	4.3×10^{-7}	6.37

(Values vary slightly between data sources: use whatever the question gives.)

MARK-SCHEME WORDING

- "Ka = $[H_3O^+][CH_3COO^-] \div [CH_3COOH]$"; ACCEPT $[H^+]$.
- DO NOT AWARD HA and A^- unless you have defined them for that acid; round or curly brackets; the charge written outside the square bracket.

Often asked as: write Ka for a named or unfamiliar acid (hexanoic, gluconic, the pyridinium ion, Tris-H^+), usually 1 mark before a calculation.

COMMON MISTAKE

- Writing the dissociation **equation** when the **expression** is asked (examiners call it "the most common error").
- Including water, or using round brackets.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Ka for **positive weak acids**, such as the pyridinium ion: $C_5H_5NH^+ \rightleftharpoons C_5H_5N + H^+$, so Ka = $[C_5H_5N][H^+] \div [C_5H_5NH^+]$ (WCH14 Jun 2024).

SPEC 14.9 pH of a weak acid

WHAT THIS SPEC POINT ASKS

Calculate the pH of a weak acid from Ka or pKa, stating the assumptions (no quadratics needed).

Method

1. If you are given pKa, find **$Ka = 10^{-pKa}$**.
2. Make **two assumptions**:
   - **$[H^+] = [A^-]$**: all the H^+ comes from the acid; the tiny amount from water is ignored.
   - **[HA]equilibrium = [HA]initial**: the acid dissociates so little that its concentration hardly changes.
3. $Ka = [H^+]^2 \div [HA]$, so **$[H^+] = \sqrt{(Ka \times [HA])}$**.
4. $pH = -\log[H^+]$.

WORKED EXAMPLE

Worked example: pH of weak acids

(a) 0.100 mol dm^{-3} ethanoic acid, $Ka = 1.7 \times 10^{-5}$ mol dm^{-3}.

$[H^+] = \sqrt{(1.7 \times 10^{-5} \times 0.100)} = 1.30 \times 10^{-3}$ mol dm^{-3}, pH = **2.88**.

(b) 0.0500 mol dm^{-3} propanoic acid, pKa = 4.87.

$Ka = 10^{-4.87} = 1.35 \times 10^{-5}$ mol dm^{-3}; $[H^+] = \sqrt{(1.35 \times 10^{-5} \times 0.0500)} = 8.21 \times 10^{-4}$; pH = **3.09**.

(c) What mass of ethanoic acid (Mr 60.0) must be dissolved to make 250 cm^3 of solution of pH 3.00?

$[H^+] = 1.00 \times 10^{-3}$; $[HA] = [H^+]^2 \div Ka = 1.00 \times 10^{-6} \div 1.7 \times 10^{-5} = 0.0588$ mol dm^{-3}

moles $= 0.0588 \times 0.250 = 0.0147$ mol; mass $= 0.0147 \times 60.0 =$ **0.88 g**.

When the assumptions fail

- The second assumption fails when the acid is **not very weak** (Ka bigger than about 10^{-3}) or very **dilute**: then a significant fraction dissociates, so [HA]eqm is noticeably **less** than [HA]initial.
- The calculated $[H^+]$ is then **too high**, so the calculated pH is **lower** than the measured pH. The error is bigger for the stronger acid.

pH of 0.0500 mol dm^{−3} ethanoic acid

1 $[\mathrm{H^+}]^2 = K_a \times [\mathrm{HA}] = 1.7 \times 10^{-5} \times 0.0500 = 8.5 \times 10^{-7}$

2 $[H^+] = \sqrt{8.5 \times 10^{-7}} = 9.22 \times 10^{-4}\ \text{mol dm}^{-3}$

3 pH = −log(9.22 × 10^{-4}) = **3.04**.

A strong acid of the same concentration would have pH 1.30. Only about 2% of the ethanoic acid molecules are dissociated.

MARK-SCHEME WORDING

- "[H^+] = √(Ka × [HA])", then −log; correct answer with no working scores full marks.
- Assumptions (any one per mark): "degree of dissociation is very small / negligible"; "[CH_3COOH]eqm = [CH_3COOH]initial"; "[H^+] = [CH_3COO^-]"; "all the hydrogen ions come from the acid / none from water".
- Questioning validity: "[HA]equilibrium is lower than [HA]initial because dissociation is significant", so the calculated pH is lower than the measured one.

Often asked as: pH from Ka/pKa (2–3 marks); state one or two assumptions; MCQ on which assumption is invalid; 6-mark comparing calculated and measured pH; mass of acid for a given pH.

COMMON MISTAKE

- Giving the **same assumption twice** in different words: give one about water ($H^+ = A^-$) and one about the degree of dissociation.
- Forgetting to convert pKa to Ka, or g dm^{-3} to mol dm^{-3} (divide by Mr).
- Blaming the dissociation of **water** for a difference between measured and calculated pH: the real reason is that dissociation of the acid is not negligible.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- An acid whose dissociation is **not** negligible: dichloroethanoic acid, Ka = 4.5 × 10^{-2} mol dm^{-3}. The simple method gives a pH that is too low (WCH14 Jun 2024).
- Working backwards to the **mass of acid** needed for a given pH (WCH14 Oct 2022). With the assumption, the mass is an underestimate, because [HA]eqm < [HA]initial means more acid is really needed.

SPEC 14.10 The ionic product of water, Kw

WHAT THIS SPEC POINT ASKS

Define Kw.

Where Kw comes from

- Water ionises very slightly: $H_2O(l) \rightleftharpoons H^+(aq) + OH^-(aq)$
- $[H_2O]$ is effectively constant, so it is absorbed into the constant:

$$\mathbf{K_w = [H^+][OH^-] = 1.00 \times 10^{-14}\ mol^2\ dm^{-6}\ at\ 298\ K}$$

- In **pure water** $[H^+] = [OH^-]$, so $[H^+] = \sqrt{Kw} = 1.00 \times 10^{-7}$ mol dm^{-3} and pH = 7.00 at 298 K.
- **Neutral** means $[H^+] = [OH^-]$, not "pH 7". Acidic: $[H^+] > [OH^-]$. Alkaline: $[OH^-] > [H^+]$.

Kw changes with temperature

- The ionisation of water is **endothermic** (it breaks bonds). Raising the temperature shifts the equilibrium **right**, so **Kw increases**.
- At higher temperature $[H^+]$ is bigger, so the **pH of pure water falls below 7**, but the water is **still neutral** because $[H^+]$ still equals $[OH^-]$.

Temperature	Kw / mol² dm⁻⁶	pH of pure water
283 K (10 °C)	2.9×10^{-15} (pKw 14.53)	7.27
298 K (25 °C)	1.00×10^{-14}	7.00
313 K (40 °C)	2.9×10^{-14}	6.77
323 K (50 °C)	5.5×10^{-14}	6.63

WORKED EXAMPLE

Worked example: pure water at 313 K

Kw at 313 K = 2.92×10^{-14} mol² dm^{-6}.

$[H^+] = \sqrt{(2.92 \times 10^{-14})} = 1.71 \times 10^{-7}$ mol dm^{-3}, pH = **6.77**. The water is neutral: $[H^+] = [OH^-]$.

MARK-SCHEME WORDING

- "Kw = $[H^+][OH^-]$" with units $mol^2\ dm^{-6}$.
- Temperature: "the dissociation of water is endothermic, so the concentration of hydrogen ions is higher at 75 °C"; "in pure water the concentrations of hydrogen ions and hydroxide ions are always equal".
- At a temperature other than 298 K use the pKw you are given: pH = pKw − pOH.

Often asked as: MCQ on the pH of pure water at 40 °C or 75 °C; "is water at pH 6.6 acidic?"; pH of NaOH at 283 K using pKw = 14.53.

COMMON MISTAKE

- Saying hot water is "acidic" because its pH is below 7.
- Using 14 for pKw when the question gives a different value.
- Including $[H_2O]$ in the Kw expression.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Neutral water that isn't pH 7** above 25 °C, and the endothermic explanation (WCH14 Jun 2025). At 100 °C Kw is about 5×10^{-13}, so pure water has pH about 6.1 (WCH14 Jan 2023).

SPEC 14.11 pH of a strong base

WHAT THIS SPEC POINT ASKS

Calculate the pH of a strong base from its concentration, using Kw or pKw.

Method

1. **$[OH^-]$** from the concentration. NaOH and KOH give $[OH^-]$ = c. $Ba(OH)_2$ and other Group 2 hydroxides give **$[OH^-] = 2c$**.
2. Either **$[H^+] = Kw \div [OH^-]$**, then pH = $-\log[H^+]$;
   or **$pOH = -\log[OH^-]$**, then **pH = pKw − pOH** (pKw = 14.00 at 298 K).

WORKED EXAMPLE

Worked example: pH of strong bases

(a) 0.0500 mol dm^{-3} NaOH at 298 K: $[H^+] = 1.00 \times 10^{-14} \div 0.0500 = 2.00 \times 10^{-13}$; pH = **12.70**.

(b) 0.0200 mol dm⁻³ $Ba(OH)_2$: $[OH^-]$ = 0.0400; pOH = 1.40; pH = 14.00 − 1.40 = **12.60**.

(c) 0.100 mol dm⁻³ NaOH at 283 K, where pKw = 14.53: pOH = 1.00, pH = **13.53**.

(d) 25.0 cm³ of 0.100 mol dm⁻³ HCl + 30.0 cm³ of 0.100 mol dm⁻³ NaOH:

excess OH^- = $3.00 \times 10^{-3} - 2.50 \times 10^{-3} = 5.0 \times 10^{-4}$ mol in 55.0 cm³

$[OH^-] = 9.09 \times 10^{-3}$ mol dm⁻³; $[H^+] = 1.00 \times 10^{-14} \div 9.09 \times 10^{-3} = 1.10 \times 10^{-12}$; pH = **11.96**.

pH of a strong base

1. Strong bases such as NaOH are fully dissociated, so 0.0500 mol dm^{-3} NaOH has $[OH^-]$ = 0.0500 mol dm^{-3}.
2. Use K_w: $[H^+] = \frac{K_w}{[OH^-]} = \frac{1.00 \times 10^{-14}}{0.0500} = 2.00 \times 10^{-13}\ \text{mol dm}^{-3}$.
3. pH = $-\log(2.00 \times 10^{-13})$ = **12.70**.

Watch for bases that release two hydroxide ions: 0.0200 mol dm^{-3} $Ba(OH)_2$ gives $[OH^-]$ = 0.0400 mol dm^{-3} and pH 12.60.

MARK-SCHEME WORDING

- "$[OH^-]$ = 0.4 mol dm⁻³" for 0.2 mol dm⁻³ $Ba(OH)_2$, pOH = 0.40, pH = 13.6.
- After a reaction that uses up OH^- (e.g. ester hydrolysis): moles NaOH − moles used, new concentration, "pH = 14 − (−log$[OH^-]$)".

Often asked as: pH of NaOH or $Ba(OH)_2$; pH of an alkali left over after a reaction or titration; MCQ at a non-standard temperature.

COMMON MISTAKE

- Taking −log$[OH^-]$ and calling it the pH.
- Forgetting the 2 for $Ba(OH)_2$, or forgetting to subtract the OH^- that has been used up.
- Not dividing by the final (combined) volume.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- pH of the excess NaOH after an ester hydrolysis: subtract the moles that reacted first; "13.99" from the original concentration scored nothing (WCH14 Jan 2020).

SPEC 14.12 pKa and pKw

WHAT THIS SPEC POINT ASKS

Define pKa and pKw.

Definitions

$$pK_a = -\log_{10} K_a \quad K_a = 10^{-pKa}$$

$$pK_w = -\log_{10} K_w = 14.00 \text{ at } 298 \text{ K}$$

- The "p" means "$-\log_{10}$ of".
- **Smaller pKa = larger Ka = stronger acid.** A difference of 1 in pKa is a factor of 10 in Ka.
- pKa values are easier to compare than Ka values with negative powers.
- At 298 K, pH + pOH = 14.00 (more generally pH + pOH = pKw).

WORKED EXAMPLE

Worked example: converting

(a) pKa = 5.25 → Ka = $10^{-5.25}$ = **5.6×10^{-6} mol dm^{-3}**.

(b) Ka = 4.5×10^{-2} mol dm^{-3} → pKa = **1.35**.

Which acid is stronger? (b), because its pKa is smaller.

MARK-SCHEME WORDING

- "inverse log(−5.25) to 2 s.f." (5.6×10^{-6}) and "−log(4.5×10^{-2}) to 3 s.f." (1.35). DO NOT AWARD 6×10^{-6} or 1.4.
- pKa and pKw have **no units**.

Often asked as: conversions inside longer calculations; rank acids by pKa.

COMMON MISTAKE

- Rounding to 1 s.f., or giving s.f. inconsistent with the data.

- Thinking a bigger pKa means a stronger acid.

SPEC 14.13 Analysing pH data: acids, bases, salts and dilution

WHAT THIS SPEC POINT ASKS

Interpret pH measurements of equimolar strong and weak acids and bases and of salts, and compare the pH of strong and weak acids after 10, 100 and 1000-fold dilution.

Equimolar acids and bases

At the **same concentration and temperature**, lower pH = stronger acid; higher pH = stronger base. Typical values for 0.100 mol dm^{-3} solutions at 298 K:

Solution (0.100 mol dm^{-3})	pH	Why
HCl	1.00	strong acid, $[H^+] = 0.100$
CH_3COOH	2.9	weak acid, ~1% dissociated
NH_3	11.1	weak base, partly protonated
NaOH	13.0	strong base, $[OH^-] = 0.100$

Salts are not all neutral

The pH of a salt solution depends on which acid and base it came from:

Salt from	Example	pH (0.1 mol dm^{-3})	Reason
strong acid + strong base	NaCl, KNO_3	7	neither ion reacts with water
weak acid + strong base	CH_3COONa	about 8.9	$CH_3COO^- + H_2O \rightleftharpoons CH_3COOH + OH^-$
strong acid + weak base	NH_4Cl	about 5.1	$NH_4^+ + H_2O \rightleftharpoons NH_3 + H_3O^+$
weak acid + weak base	CH_3COONH_4	about 7	both ions react, about equally

- The anion of a weak acid is a weak **base**; the cation of a weak base is a weak **acid**. That's why the equivalence point of a titration is not always at pH 7 (14.15).

Dilution: strong v weak acid

Dilution of 0.100 mol dm^{-3}	×1	×10	×100	×1000
HCl, pH	1.00	2.00	3.00	4.00
CH_3COOH, pH	2.88	3.38	3.88	about 4.4

- **Strong acid**: each tenfold dilution divides [H^+] by 10, so the pH rises by **1 unit**.
- **Weak acid**: the pH rises by only about **0.5 units** per tenfold dilution. Two ways to see why:
  - [H^+] = $\sqrt{(Ka \times c)}$, so dividing c by 10 divides [H^+] by $\sqrt{10} \approx 3.2$, and $\log 3.2 \approx 0.5$.
  - Le Chatelier: diluting shifts $HA \rightleftharpoons H^+ + A^-$ to the right (more particles side), so the acid **dissociates more** and partly makes up for the dilution.
- At great dilution the weak acid's dissociation is no longer small (about 12% at 0.0001 mol dm^{-3}), so the simple formula slightly underestimates the pH.
- Diluting an acid can never take its pH above 7: at around 10^{-8} mol dm^{-3} the H^+ from water dominates and the pH just approaches 7.

MARK-SCHEME WORDING

- Dilution (4 points): HCl "fully dissociated"; "a decrease in concentration by a factor of 10 decreases [H^+] by a factor of 10" so pH **increases** by 1; ethanoic acid "not fully dissociated (so its pH is much higher)"; "when diluted, ethanoic acid dissociates more (so the pH does not change so much)". IGNORE "pH increases by 0.5" on its own.
- Ammonium chloride: "$NH_4^+ + H_2O \rightleftharpoons NH_3 + H_3O^+$".

Often asked as: explain the different pH changes on dilution; rank the pH of several solutions (MCQ); explain why a salt solution is acidic or alkaline.

COMMON MISTAKE

- Saying dilution "decreases the pH". Diluting an acid **raises** its pH.
- Writing NH_4Cl, rather than the NH_4^+ ion, in the hydrolysis equation ("the majority used ammonium chloride").
- Comparing acids of **different** concentrations as if pH alone showed strength.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Salt solutions with pH ≠ 7**, with the hydrolysis equation for the ion that reacts (WCH14 Jun 2022).

SPEC 14.14 Ka from experimental data

WHAT THIS SPEC POINT ASKS

Calculate Ka of a weak acid from the pH of a solution made from a known mass of the acid.

Method

1. **Concentration** of the acid: moles = mass ÷ Mr; c = moles ÷ volume in dm³.
2. **$[H^+] = 10^{-pH}$** from the measured pH.
3. Assume **$[A^-] = [H^+]$** and **$[HA] \approx c$** (dissociation small).
4. **$K_a = [H^+]^2 \div c$**, units mol dm^{-3}. Then pKa = −log Ka if asked.
5. If dissociation is not small, the better value uses **$[HA] = c - [H^+]$**.

WORKED EXAMPLE

Worked example: Ka from mass and pH

0.600 g of ethanoic acid (Mr 60.0) is dissolved in water and made up to 250 cm³. The pH is 3.08.

moles = 0.600 ÷ 60.0 = 0.0100 mol; c = 0.0100 ÷ 0.250 = 0.0400 mol dm^{-3}

$[H^+] = 10^{-3.08} = 8.32 \times 10^{-4}$ mol dm^{-3}

$K_a = (8.32 \times 10^{-4})^2 \div 0.0400 =$ **1.7×10^{-5} mol dm^{-3}** (pKa = 4.76)

Percentage dissociated = $8.32 \times 10^{-4} \div 0.0400 \times 100 = 2.1\%$, so the assumption is fine.

WORKED EXAMPLE

Worked example: checking the assumption

0.244 g of benzoic acid (Mr 122) in 100 cm³ of solution has pH 2.94.

c = 0.00200 mol ÷ 0.100 dm³ = 0.0200 mol dm^{-3}; $[H^+] = 10^{-2.94} = 1.15 \times 10^{-3}$ mol dm^{-3}

With [HA] = c: $K_a = (1.15 \times 10^{-3})^2 \div 0.0200 =$ **6.6×10^{-5} mol dm^{-3}**

Percentage dissociated = 5.7%. Using [HA] = 0.0200 − 0.00115 = 0.01885 gives $K_a = 7.0 \times 10^{-5}$ mol dm^{-3}, a better value.

K_a from a measured pH

1. 0.200 mol dm^{-3} propanoic acid has pH 2.79. Find K_a and pK_a.
2. $[H^+] = 10^{-2.79} = 1.62 \times 10^{-3}$ mol dm^{-3} = $[A^-]$.
3. $K_a = \frac{(1.62 \times 10^{-3})^2}{0.200} = 1.32 \times 10^{-5}\ \text{mol dm}^{-3}$, so $pK_a = 4.88$.

MARK-SCHEME WORDING

- Steps credited: concentration of acid from mass ÷ Mr ÷ volume; $[H^+]$ from pH; Ka expression; Ka with units.
- Phosphoric acid type: "$[H^+] = 10^{-1.2} = 0.0631$", "assume $[H_2PO_4^-] = [H_3O^+]$", Ka_1 with units.

Often asked as: Ka from a mass and a pH; Ka of a polyprotic acid's first dissociation; part of Core Practical 11 questions.

COMMON MISTAKE

- Using the mass concentration (g dm^{-3}) as if it were mol dm^{-3}.
- Using the Mr of the **ion** instead of the acid or salt.
- Using the initial concentration without comment when the acid is quite strong: the best answers subtract $[H^+]$.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- For a relatively strong acid such as H_3PO_4, the best answer uses the equilibrium concentration $[H_3PO_4] = c - [H^+]$; "only a very few" did this (WCH14 Oct 2020).

SPEC 14.15 Titration curves

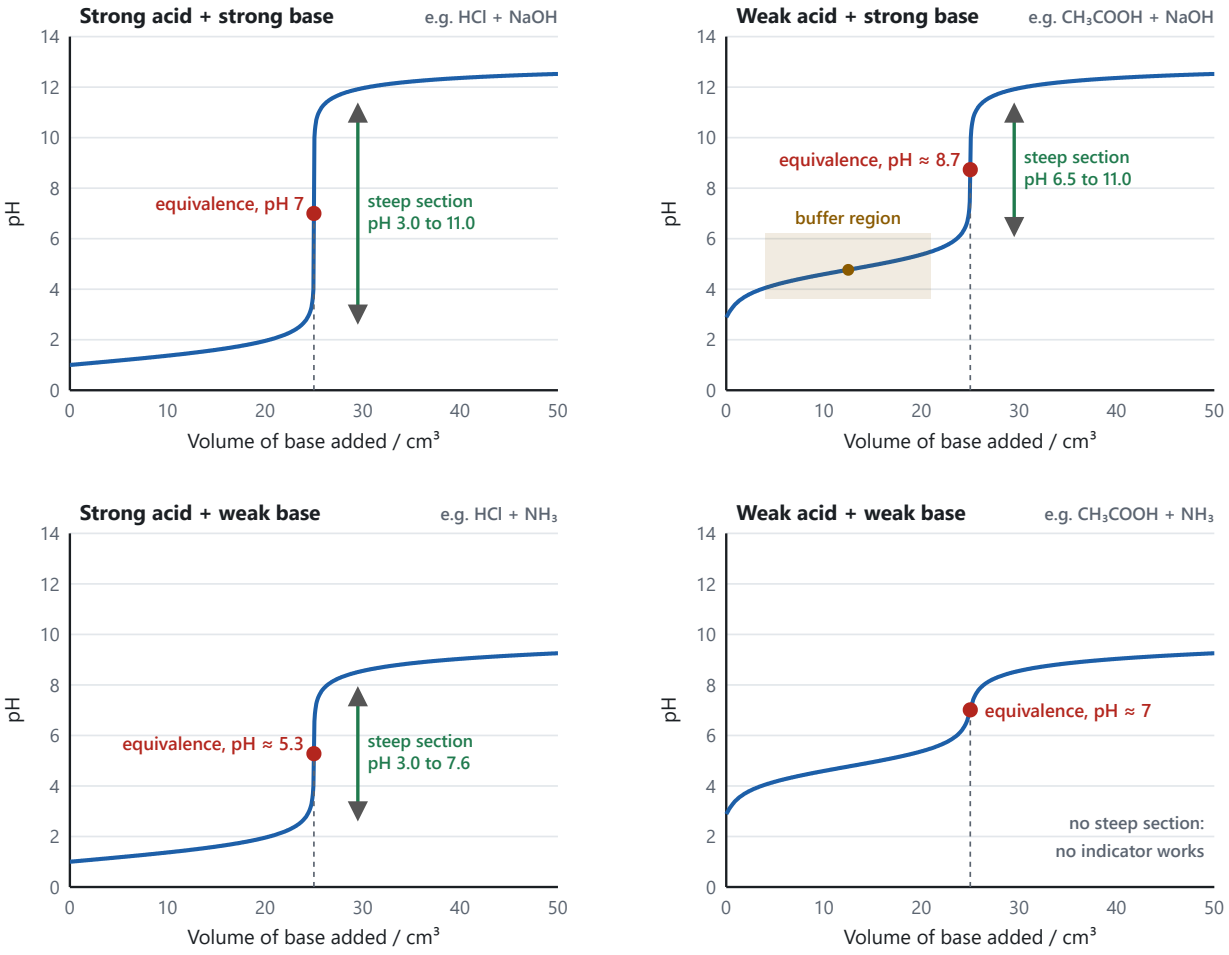
WHAT THIS SPEC POINT ASKS

Draw and interpret pH titration curves for all combinations of strong and weak monoprotic acids and bases, and apply the same ideas to diprotic acids and bases.

Reading a pH curve

A pH curve plots **pH (y-axis)** against **volume of solution added (x-axis)**. Things to read off:

- **Starting pH**: tells you what is in the flask (strong acid about 1; weak acid about 3).
- **Equivalence point**: where the moles of acid and base match the equation. It is the **middle of the steep (vertical) section**. Its volume gives you the concentration; its pH tells you which salt is present.
- **Steep section**: near equivalence, almost no H^+ (or OH^-) is left, so one drop of titrant is a large fraction of what remains and the pH jumps several units.
- **Final pH**: approaches (but stays below) the pH of the solution being added, because it is diluted.



Calculated curves: 25.0 cm³ of 0.100 mol dm⁻³ acid with 0.100 mol dm⁻³ base. The steep section is shown from 24.5 to 25.5 cm³ of base added.

The four combinations

Acid + base	Start pH	Equivalence pH	Steep section	End pH
Strong + strong (HCl + NaOH)	about 1	**7**	about 3 to 11	about 12–13
Weak + strong (CH_3COOH + NaOH)	about 3	**above 7** (about 8.7)	about 6.5 to 11	about 12–13
Strong + weak (HCl + NH_3)	about 1	**below 7** (about 5.3)	about 3 to 7.5	about 9–10
Weak + weak (CH_3COOH + NH_3)	about 3	about 7	**none**	about 9–10

- **Why the equivalence pH differs**: at equivalence only the salt is present. CH_3COO^- is a weak base (pH > 7); NH_4^+ is a weak acid (pH < 7). See the salt table in 14.13.
- **Weak acid + strong base** has an initial small rise, then a gently sloping **buffer region** (14.21), then the steep section.

Why the equivalence point of a weak acid is above 7 (extension)

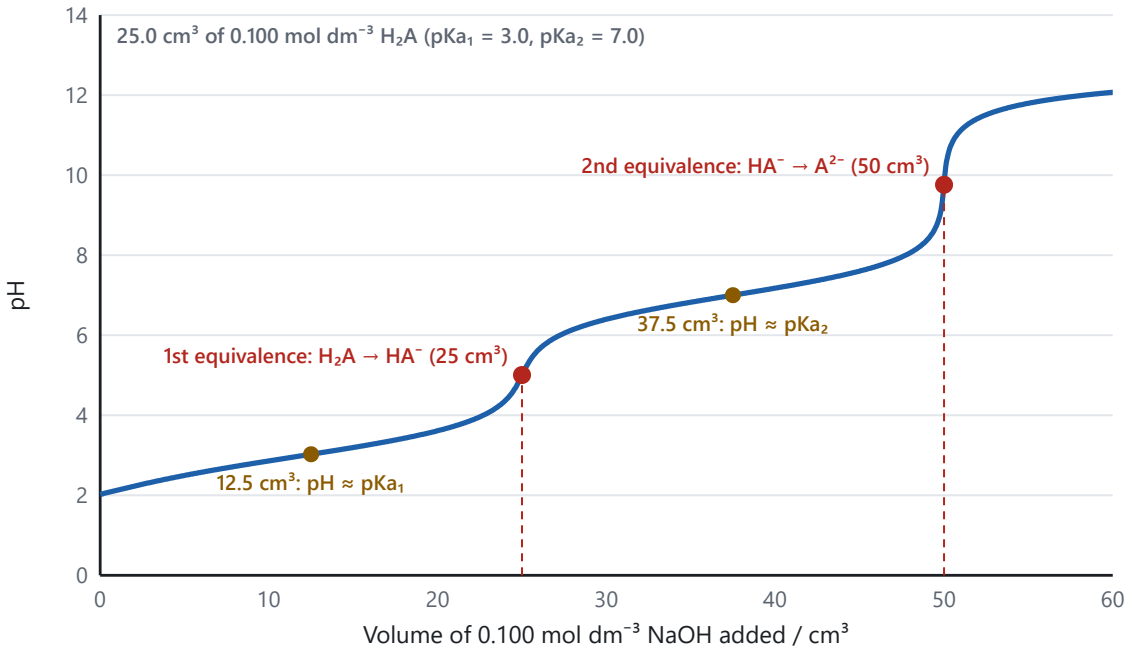
At equivalence, 25.0 cm^3 of 0.100 mol dm^{-3} CH_3COOH + 25.0 cm^3 of NaOH gives 0.0500 mol dm^{-3} CH_3COO^-, which reacts slightly with water: $CH_3COO^- + H_2O \rightleftharpoons CH_3COOH + OH^-$. Using $K_b = K_w \div K_a = 5.9 \times 10^{-10}$: $[OH^-] = \sqrt{(K_b \times c)} = 5.4 \times 10^{-6}$ mol dm^{-3}, so pOH = 5.27 and pH = **8.73**. You won't be asked for this calculation, but you must be able to say that "the salt of a weak acid is basic".

Acid added to alkali

If the **acid is in the burette** and the alkali in the flask, the curve is the same shape **upside down**: it starts high and falls. Check which solution is being added before you sketch.

Diprotic acids and bases

- A **diprotic acid** H_2A gives up two protons in turn: $H_2A + OH^- \rightarrow HA^- + H_2O$, then $HA^- + OH^- \rightarrow A^{2-} + H_2O$.
- If the two pKa values are well apart, the curve has **two steep sections**: the second equivalence volume is **twice** the first.
- Half-way to each equivalence point, pH ≈ pK_{a1} and pH ≈ pK_{a2}.
- A **diprotic base** behaves the same way. Sodium carbonate titrated with HCl shows two steps: $CO_3^{2-} \rightarrow HCO_3^-$ (near pH 8.3) and then $HCO_3^- \rightarrow H_2CO_3$ ($CO_2 + H_2O$, near pH 3.7).
- With a strong diprotic acid such as H_2SO_4 titrated with NaOH, both protons are removed together and there is one steep section, at the volume for 2 mol NaOH per mol H_2SO_4.



A diprotic weak acid with NaOH: two steep sections, at 25 and 50 cm³. At 12.5 and 37.5 cm³ the pH is close to pK_{a1} and pK_{a2}.

WORKED EXAMPLE

Worked example: reading a curve

25.0 cm³ of propanoic acid is titrated with 0.100 mol dm⁻³ NaOH. The steep section is centred on 20.0 cm³. At 10.0 cm³ the pH is 4.87.

(a) Concentration of the acid: moles NaOH = 0.100 × 0.0200 = 2.00×10^{-3} mol = moles acid (1 : 1). c = $2.00 \times 10^{-3} \div 0.0250$ = **0.0800 mol dm⁻³**.

(b) Ka: 10.0 cm³ is half-way to equivalence, so pH = pKa = 4.87. Ka = $10^{-4.87}$ = **1.3×10^{-5} mol dm⁻³**.

(c) The acid is weak: the start pH is about 3 and the equivalence point is above 7.

MARK-SCHEME WORDING

- Sketch of weak acid + strong base (any three): "start pH between 2 and 4 and finish at 12–14"; "vertical section of at least 3 pH units starting at or above 6 and finishing at or below 11"; "S-shaped curve with equivalence at the right volume".
- Diprotic: "vertical section at 25 cm³ with height 1–4 pH units"; a second vertical section at 50 cm³.
- Weak acid + strong base: equivalence pH is above 7 because "the salt of a weak acid is basic".

Often asked as: sketch a curve on a grid; pick which curve matches a pair of solutions; read the equivalence volume; sketch a diprotic curve; calculate the pH at a point on the curve.

COMMON MISTAKE

- Curves that don't start or end at a sensible pH, or a steep section that isn't vertical.
- Reading the equivalence volume as the last volume plotted (e.g. 46 cm^3) instead of the middle of the steep section (22 cm^3).
- Drawing the curve going up when acid is added to alkali.
- Only one steep section for a diprotic acid whose pKa values are well apart.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **pH at a point on the curve** by moles (e.g. after excess alkali is added), and showing why the curve levels off below the pH of the alkali added: it is diluted and partly neutralised (WCH14 Oct 2023).
- Choosing the right curve for **acid added to ammonia** (WCH14 Jun 2022).

SPEC 14.16 Choosing an indicator

WHAT THIS SPEC POINT ASKS

Choose a suitable indicator for a titration from its pH curve and indicator data.

How indicators work

- An indicator is a **weak acid** whose undissociated form and conjugate base have different colours: $HIn \rightleftharpoons H^+ + In^-$.
- In acid, the equilibrium lies left (colour of HIn); in alkali, right (colour of In^-).
- The colour changes over a range of about **pKIn ± 1**. At the **end point**, [HIn] = [In^-] and the solution shows the **mixed colour** (orange for methyl orange, green for bromothymol blue, pale pink for phenolphthalein).

The rule

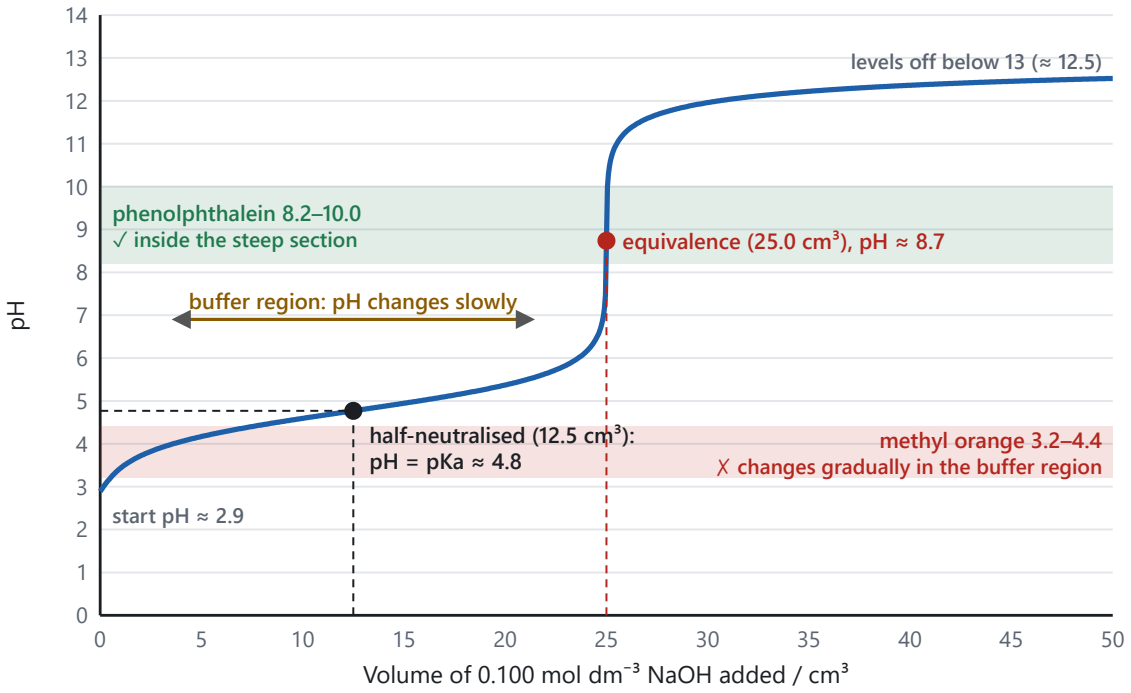
Choose an indicator whose whole colour-change range lies within the steep (vertical) section of the curve. Then one drop of titrant at the equivalence point changes the colour completely.

Indicator (Data Booklet)	pH range	Colour: acid → alkali
methyl orange	3.2–4.4	red → yellow
bromophenol blue	2.8–4.6	yellow → blue

Indicator (Data Booklet)	pH range	Colour: acid → alkali
bromocresol green	3.8–5.4	yellow → blue
methyl red	4.2–6.3	red → yellow
bromothymol blue	6.0–7.6	yellow → blue
phenol red	6.8–8.4	yellow → red
thymol blue (base)	8.0–9.6	yellow → blue
phenolphthalein	8.2–10.0	colourless → pink

(Check the exact ranges in your own Data Booklet.)

Titration	Steep section	Suitable indicator
strong acid + strong base	about 3–11	**methyl orange or phenolphthalein** (or anything in between)
weak acid + strong base	about 6.5–11	**phenolphthalein** (also thymol blue, phenol red)
strong acid + weak base	about 3–7.5	**methyl orange** (also methyl red, bromocresol green)
weak acid + weak base	none	**no indicator** works; use a pH meter



Ethanoic acid (25.0 cm³, 0.100 mol dm⁻³) with 0.100 mol dm⁻³ NaOH. Phenolphthalein changes inside the steep section; methyl orange would change slowly in the buffer region, long before equivalence.

MARK-SCHEME WORDING

- "phenolphthalein / bromothymol blue / phenol red" "because the indicator pH range / pKIn ± 1 lies (completely) within the vertical section of the graph". Two points: name, then reason.
- End-point colours: methyl orange **orange**; bromophenol blue, bromothymol blue, bromocresol green **green**; phenol red **orange**; phenolphthalein **(pale) pink**.
- Methyl orange in a weak acid titration "starts red and turns orange across the buffer region", so the end point can't be seen sharply.

Often asked as: choose from a Data Booklet list and justify; explain why an indicator is unsuitable; state the colour at the end point; colour changes during a titration or on dilution.

COMMON MISTAKE

- Saying the indicator "changes colour at the equivalence point": the mark is for its **range lying within the vertical section**.
- Quoting the Data Booklet colours (e.g. "red to yellow") as the **end-point** colour: the end point is the **in-between** colour.
- Describing whether an indicator is suitable when the question asked you to **describe the colours** seen.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Indicators with two ranges, e.g. thymol blue is listed twice ("acid" 1.2–2.8 and "base" 8.0–9.6); use the right one (WCH14 Jun 2025).
- A diprotic titration needs **one indicator per equivalence point**, and two indicators in one flask is a bad idea because "the colours will mix" (WCH14 Jan 2025).

SPEC 14.17 What a buffer solution is

WHAT THIS SPEC POINT ASKS

Know what a buffer solution is.

Definition

A **buffer solution resists changes in pH** when **small** amounts of **acid or alkali** are added (or when it is diluted).

- "Resists" does not mean "prevents": the pH changes a little.
- Add a lot of acid or alkali and the buffer is overwhelmed.

What buffers are made of

Type	Made from	Example	pH
Acidic buffer	a weak acid + a salt of that weak acid (its conjugate base)	CH_3COOH + CH_3COONa	below 7
	or: excess weak acid partly neutralised by a strong base	2 mol CH_3COOH + 1 mol NaOH	
Basic buffer	a weak base + a salt of that weak base (its conjugate acid)	NH_3 + NH_4Cl	above 7
	or: excess weak base partly neutralised by a strong acid	2 mol NH_3 + 1 mol HCl	

- A strong acid and its salt (HCl + NaCl) is **not** a buffer: Cl^- is too weak a base to take up added H^+.
- If the strong acid or base is **in excess** when you mix, there is no weak acid/conjugate base pair left, so no buffer.

MARK-SCHEME WORDING

- "a buffer resists change in pH on the addition of small amounts of acid and / or of alkali"; ACCEPT "maintains a fairly / nearly constant pH".
- Both "small" and "acid and alkali" are needed.

Often asked as: define a buffer (1–2 marks); MCQ "which mixture forms a buffer?".

COMMON MISTAKE

- Leaving out "small amounts", or saying the pH "does not change".
- Picking a strong acid + its salt, or a mixture with the strong reagent in excess, as a buffer.

SPEC 14.18 How a buffer works

WHAT THIS SPEC POINT ASKS

Explain the action of a buffer solution.

The acidic buffer CH_3COOH / CH_3COO^-

$CH_3COOH(aq) \rightleftharpoons CH_3COO^-(aq) + H^+(aq)$

- The buffer contains a **large reservoir** of both **CH_3COOH** (from the acid, barely dissociated) and **CH_3COO^-** (from the fully dissociated salt).
- **Add a little acid (H^+)**: the H^+ reacts with the ethanoate ions:
  $CH_3COO^- + H^+ \rightarrow CH_3COOH$
  The equilibrium shifts **left**; almost all the added H^+ is removed.
- **Add a little alkali (OH^-)**: OH^- reacts with H^+ (and with ethanoic acid directly):
  $H^+ + OH^- \rightarrow H_2O$, or $CH_3COOH + OH^- \rightarrow CH_3COO^- + H_2O$
  The equilibrium shifts **right**: more CH_3COOH dissociates to replace the H^+ used.
- **Why the pH hardly changes**: $[H^+] = Ka \times [HA] \div [A^-]$. Because [HA] and $[A^-]$ are both large, a small addition changes the **ratio [HA]/[A^-]** only slightly, so $[H^+]$ stays almost constant.

The basic buffer NH_3 / NH_4^+

$NH_3(aq) + H^+(aq) \rightleftharpoons NH_4^+(aq)$

- Add H^+: $NH_3 + H^+ \rightarrow NH_4^+$ (large reservoir of NH_3 removes it).
- Add OH^-: $NH_4^+ + OH^- \rightarrow NH_3 + H_2O$ (large reservoir of NH_4^+ removes it).

Buffer capacity

- A buffer only works while **both** reservoirs last. Once most of the A^- (or HA) has been used up, the ratio changes fast and so does the pH.
- More concentrated buffers have a bigger capacity. A buffer works best when [HA] ≈ [A^-], i.e. pH ≈ pKa (within about ±1).

MARK-SCHEME WORDING

- M1 "(large) reservoir of CH_3COOH and CH_3COO^-"; M2 "OH^- ions react with CH_3COOH / H^+ ions"; M3 an equation, "$CH_3COOH + OH^- \rightarrow CH_3COO^- + H_2O$"; M4 "$H^+$ ions react with CH_3COO^-" with "$CH_3COO^- + H^+ \rightarrow CH_3COOH$".
- "base and conjugate acid present in high concentrations"; "the ratio [base]/[acid] only changes very slightly so the pH changes only slightly".

Often asked as: explain with equations how a named buffer works (3–5 marks); write the equation for the reaction with added acid or alkali.

COMMON MISTAKE

- Writing equations with no words to explain them, or only giving the definition.
- Adding H^+ to the wrong member of the pair (H^+ reacts with the **conjugate base**).
- Forgetting to mention the **large reservoir** of the weak acid.
- Writing a species that isn't in the buffer (e.g. H_2SO_4 in an HSO_4^-/SO_4^{2-} buffer).

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- Explaining buffer action **through the ratio** [salt]/[acid], not just "resists change" (WCH14 Jan 2026).
- Buffers made from less familiar pairs: **HSO_4^- / SO_4^{2-}** (WCH14 Jan 2021), **$H_2PO_4^-$ / HPO_4^{2-}** and the base **Tris** with its conjugate acid (WCH14 Jun 2023).

SPEC 14.19 Calculating the pH of a buffer

WHAT THIS SPEC POINT ASKS

Calculate the pH of a buffer solution from given data.

The method (Ka route)

1. Write Ka = $[H^+][A^-] \div [HA]$ and rearrange:

$$[H^+] = K_a \times [HA] \div [A^-]$$

2. Assume **[HA] = concentration of the acid added** and **$[A^-]$ = concentration of the salt added** (the acid barely dissociates and the salt is fully dissociated).
3. pH = −log$[H^+]$.

- Because both are in the same volume, you can use **moles** instead of concentrations: the volume cancels.
- The equivalent log form (Henderson–Hasselbalch) is: **pH = pKa + log($[A^-]$ ÷ [HA])**. Examiners repeatedly see it used with the ratio **upside down**, so the Ka route is safer.
- A buffer is **not** a weak acid on its own: $[H^+] \neq [A^-]$ here.
- **Basic buffers** use Ka of the conjugate acid in exactly the same way. 0.100 mol dm^{-3} NH_3 with 0.200 mol dm^{-3} NH_4Cl, Ka(NH_4^+) = 5.6×10^{-10}: $[H^+] = 5.6 \times 10^{-10} \times 0.200 \div 0.100 = 1.12 \times 10^{-9}$ mol dm^{-3}, pH = 8.95.

WORKED EXAMPLE

Worked example: acid + salt

4.10 g of sodium ethanoate (Mr 82.0) is dissolved in 250 cm^3 of 0.300 mol dm^{-3} ethanoic acid. Ka = 1.7×10^{-5} mol dm^{-3}.

$[CH_3COO^-]$ = (4.10 ÷ 82.0) ÷ 0.250 = 0.200 mol dm^{-3}; $[CH_3COOH]$ = 0.300 mol dm^{-3}

$[H^+] = 1.7 \times 10^{-5} \times 0.300 \div 0.200 = 2.55 \times 10^{-5}$ mol dm^{-3}; pH = **4.59**.

WORKED EXAMPLE

Worked example: buffer made by partial neutralisation

50.0 cm^3 of 0.200 mol dm^{-3} CH_3COOH is mixed with 30.0 cm^3 of 0.200 mol dm^{-3} NaOH.

moles CH_3COOH = 0.0100; moles NaOH = 0.00600

$CH_3COOH + OH^- \rightarrow CH_3COO^- + H_2O$, so after reaction: CH_3COOH left = 0.00400 mol; CH_3COO^- formed = 0.00600 mol

$[H^+] = 1.7 \times 10^{-5} \times 0.00400 \div 0.00600 = 1.13 \times 10^{-5}$ mol dm^{-3}; pH = **4.95**.

WORKED EXAMPLE

Worked example: adding acid or alkali to a buffer

A buffer contains 0.0100 mol CH_3COOH and 0.0100 mol CH_3COO^- in 100 cm^3. Its pH = pKa = 4.77.

(a) Add 1.00 cm^3 of 1.00 mol dm^{-3} HCl (0.00100 mol H^+). The H^+ turns CH_3COO^- into CH_3COOH:

CH_3COOH = 0.0110 mol; CH_3COO^- = 0.0090 mol

$[H^+] = 1.7 \times 10^{-5} \times 0.0110 \div 0.0090 = 2.08 \times 10^{-5}$; pH = **4.68** (a fall of only 0.09).

(b) Instead add 0.00100 mol NaOH. The OH^- turns CH_3COOH into CH_3COO^-:

CH_3COOH = 0.0090 mol; CH_3COO^- = 0.0110 mol; pH = **4.86**.

(c) Compare: the same HCl added to 100 cm³ of pure water gives $[H^+] = 0.00100 \div 0.101 = 9.9 \times 10^{-3}$, pH = **2.00**, a fall of 5 units.

Buffer made by mixing acid and salt

1. 50.0 cm^3 of 0.200 mol dm^{-3} ethanoic acid is mixed with 50.0 cm^3 of 0.100 mol dm^{-3} sodium ethanoate. $K_a = 1.7 \times 10^{-5}$ mol dm^{-3}.
2. Moles of acid = 0.0500 × 0.200 = 0.0100 mol. Moles of ethanoate = 0.0500 × 0.100 = 0.00500 mol.
3. $[H^+] = 1.7 \times 10^{-5} \times \frac{0.0100}{0.00500} = 3.4 \times 10^{-5}\ \text{mol dm}^{-3}$, so pH = **4.47**.

Buffer made by partly neutralising a weak acid

1. 50.0 cm^3 of 0.400 mol dm^{-3} ethanoic acid is mixed with 30.0 cm^3 of 0.200 mol dm^{-3} NaOH. Find the pH.
2. Moles of acid = 0.0200 mol; moles of NaOH = 0.00600 mol. The NaOH reacts completely: $CH_3COOH + NaOH \rightarrow CH_3COONa + H_2O$.
3. Left over: 0.0200 − 0.00600 = 0.0140 mol of acid; formed: 0.00600 mol of ethanoate.
4. $[H^+] = 1.7 \times 10^{-5} \times \frac{0.0140}{0.00600} = 3.97 \times 10^{-5}\ \text{mol dm}^{-3}$, so pH = **4.40**.

Adding acid to a buffer

1. 1.00 dm^3 of buffer contains 0.100 mol of ethanoic acid and 0.100 mol of sodium ethanoate, so its pH = pK_a = 4.77. Now 0.0100 mol of HCl is added.
2. The H^+ turns ethanoate into ethanoic acid: acid becomes 0.110 mol, ethanoate becomes 0.0900 mol.

3 $[H^+] = 1.7 \times 10^{-5} \times \frac{0.110}{0.0900} = 2.08 \times 10^{-5}\ \text{mol dm}^{-3}$, so pH = **4.68**, a fall of only 0.09.

The same acid added to 1.00 dm^3 of pure water would take the pH from 7.00 to 2.00.

MARK-SCHEME WORDING

- "[H⁺] = Ka × [HA] ÷ [A⁻]" then pH; the moles route gives the same answer; Henderson–Hasselbalch is allowed if quoted and used correctly. "If the final answer is above pH 7 do not award" (acidic buffer).
- Partial neutralisation: use the moles **remaining** after reaction, not the starting amounts.
- Adding acid: recalculate **both** members of the pair.

Often asked as: pH of a buffer from masses or volumes (3–5 marks); pH after mixing a weak acid with less than its equivalent of NaOH; compare the pH change on adding acid to the buffer and to water; MCQ with distractors that invert the ratio or use ln.

COMMON MISTAKE

- Putting [H⁺] = [salt] or [H⁺] = [A⁻] in a buffer.
- Inverting the ratio in pH = pKa + log([A⁻]/[HA]) or getting the sign wrong.
- Using the starting moles of acid when some has been neutralised.
- Treating the buffer as a plain weak acid.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Partial neutralisation** buffers (WCH14 Jan 2020): when the weak acid is in twofold excess, half is neutralised and pH = pKa (WCH14 Oct 2025).
- A **phosphate** buffer, $H_2PO_4^-$ / HPO_4^{2-}, with pH about 7.2 (WCH14 Jan 2026).

SPEC 14.20 Preparing a buffer of a given pH

WHAT THIS SPEC POINT ASKS

Calculate the concentrations needed to prepare a buffer with a chosen pH.

Method

1. Choose a weak acid with **pKa close to the target pH** (within about 1).
2. **$[H^+] = 10^{-pH}$** for the target pH.
3. Rearrange Ka: **$[A^-] \div [HA] = Ka \div [H^+]$**.
4. Fix one concentration (or the volume) and calculate the other; convert to moles or mass if asked.

WORKED EXAMPLE

Worked example: mass of salt for a target pH

Make 500 cm^3 of a pH 5.00 buffer using 0.100 mol dm^{-3} ethanoic acid ($K_a = 1.7 \times 10^{-5}$). What mass of sodium ethanoate (Mr 82.0) is needed?

$[H^+] = 1.0 \times 10^{-5}$ mol dm^{-3}

$[A^-] = Ka \times [HA] \div [H^+] = 1.7 \times 10^{-5} \times 0.100 \div 1.0 \times 10^{-5} = 0.170$ mol dm^{-3}

moles in 500 cm^3 = 0.170 × 0.500 = 0.0850 mol; mass = 0.0850 × 82.0 = **6.97 g**.

WORKED EXAMPLE

Worked example: mixing volumes

You have 0.100 mol dm^{-3} ethanoic acid and 0.100 mol dm^{-3} sodium ethanoate. What volumes give 100 cm^3 of the pH 5.00 buffer?

Needed ratio $[A^-]$: [HA] = 1.7 : 1. The solutions have equal concentrations, so the volumes must be in the same ratio.

Volume of acid = 100 ÷ 2.7 = **37.0 cm^3**; volume of salt solution = **63.0 cm^3**.

MARK-SCHEME WORDING

- $[H^+]$ from pH; rearranged Ka; [salt]; scaled to the volume to give moles; × Mr for mass.
- Ratio questions: give the ratio **in the order asked** (acid : salt or salt : acid).

Often asked as: moles or mass of salt to add for a target pH; acid : salt ratio for a given pH.

COMMON MISTAKE

- Leaving out the scaling step (concentration → moles in the actual volume).
- Giving salt : acid when acid : salt was asked.
- Not noticing that the answer makes no sense (e.g. an "acidic" buffer needing a pH above 7).

SPEC 14.21 Using titration curves: buffer action and Ka

WHAT THIS SPEC POINT ASKS

Use a weak acid–strong base (or strong acid–weak base) titration curve to show buffer action and to find Ka from the pH at half-neutralisation.

The buffer region

- In a weak acid–strong base titration, after the first few cm³ the flask contains both unreacted **HA** and the **A^-** made by the reaction. That mixture is a **buffer**.
- So the curve is **flat and gently sloping** over a wide range of volume: adding OH^- changes the pH only slowly. This is the **buffer region**.

Half-neutralisation: pH = pKa

- When **half** the acid has been neutralised (half the equivalence volume), **[HA] = [A^-]**.
- Ka = [H^+][A^-] ÷ [HA], and the two concentrations cancel, so **Ka = [H^+]** and **pH = pKa**.
- So: find the equivalence volume, halve it, read the pH there, and **$Ka = 10^{-pH}$**.
- For a weak **base** titrated with strong acid, the pH at half-neutralisation equals the pKa of the **conjugate acid** (e.g. 9.25 for NH_4^+).
- For a diprotic acid, read pH at ¼ and ¾ of the second equivalence volume: these give pKa_1 and pKa_2.

WORKED EXAMPLE

Worked example: Ka from the curve

A weak acid is titrated with NaOH. The equivalence point is at 18.0 cm³. At 9.0 cm³ the pH is 4.20.

At 9.0 cm³, half the acid is neutralised, so [HA] = [A^-] and pH = pKa = 4.20.

Ka = $10^{-4.20}$ = **6.3×10^{-5} mol dm^{-3}**. (Show the 9.0 cm³ line and the pH reading on the graph.)

MARK-SCHEME WORDING

- "(when half the acid has been neutralised) [CH_3CH_2COOH] = [$CH_3CH_2COO^-$]" and "pH at half neutralisation = pKa". ALLOW in words.
- Read-off tolerance: e.g. "pH 3.8 ± 0.1", then $Ka = 10^{-pH}$; TE from your equivalence volume.
- Buffer region: "this region is a buffer"; "large reservoir of undissociated acid (and its anion)"; added OH^- reacts with H^+ and more acid dissociates.

Often asked as: find Ka (or pKa) from a curve; identify and explain the flat region; Ka_1 and Ka_2 of a diprotic acid.

COMMON MISTAKE

- Not realising that the pH at half-neutralisation gives pKa directly.
- Reading the pH at the equivalence point or at the end of the curve (e.g. 8 or 13) instead.
- Giving Ka to 1 s.f. ("rarely acceptable") or leaving no working on the graph.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- **Ka_1 and Ka_2 of a diprotic acid** from the two half-equivalence points (WCH14 Jan 2025).
- Doing a full buffer calculation at the half-neutralisation point gives the same answer and scores (WCH14 Oct 2023).

SPEC 14.22 Buffers in living things and food

WHAT THIS SPEC POINT ASKS

Understand why buffers matter in cells and blood (H_2CO_3/HCO_3^-) and in food (preventing spoilage from pH changes caused by bacteria or fungi).

Blood: the carbonic acid–hydrogencarbonate buffer

- Blood pH must stay within **7.35–7.45**. Enzymes and proteins (e.g. haemoglobin) only work properly in a narrow pH range.
- Two linked equilibria:
  $CO_2(aq) + H_2O(l) \rightleftharpoons H_2CO_3(aq)$
  $H_2CO_3(aq) \rightleftharpoons H^+(aq) + HCO_3^-(aq)$
- **Extra H^+** (e.g. lactic acid during exercise, or extra CO_2 from respiration): the large reservoir of **HCO_3^-** removes it, $HCO_3^- + H^+ \rightarrow H_2CO_3$; the H_2CO_3 breaks down to CO_2, which is **exhaled**.
- **Loss of H^+** (or OH^- added): H_2CO_3 dissociates to replace it, and more CO_2 dissolves to make more H_2CO_3.
- The ratio $[HCO_3^-] : [H_2CO_3]$ stays nearly constant, so $[H^+]$ does too. The pH can still shift slightly (it falls a little during exercise).
- Inside **cells**, phosphate ($H_2PO_4^-$ / HPO_4^{2-}) and proteins act as buffers too.

WORKED EXAMPLE

Worked example: the ratio in blood

Ka of H_2CO_3 = 4.3×10^{-7} mol dm^{-3}. What ratio $[HCO_3^-]$: $[H_2CO_3]$ gives pH 7.40?

$[H^+] = 10^{-7.40} = 3.98 \times 10^{-8}$ mol dm^{-3}

$[HCO_3^-] \div [H_2CO_3] = Ka \div [H^+] = 4.3 \times 10^{-7} \div 3.98 \times 10^{-8} =$ **10.8 : 1**.

There is far more HCO_3^- than H_2CO_3, so blood is better at mopping up added acid than added alkali, which suits the body (metabolism produces acids).

Food

- Bacteria and fungi growing in food produce acids (or bases), which change the pH and spoil flavour and texture; a changed pH can also help some microbes grow.
- Adding a buffer such as **citric acid + sodium citrate** keeps the pH almost constant, which helps **prevent deterioration**. Buffers are also used in drinks, jams and some medicines.

MARK-SCHEME WORDING

- Blood (5 marks): equations for $CO_2 + H_2O \rightleftharpoons H_2CO_3$ and $H_2CO_3 \rightleftharpoons H^+ + HCO_3^-$; "addition of CO_2 (from respiration)"; "removal of CO_2 (by exhalation)"; "there is a large excess of hydrogencarbonate"; "the ratio of salt / acid remains (almost) the same".
- "(large concentration of) HCO_3^- reacts with (extra) H^+ ions".
- Food: "(sodium citrate) forms a buffer (with citric acid) which maintains an (almost) constant pH to prevent deterioration (due to pH change caused by bacteria or fungal activity)".
- Why pH matters: "pH is important for enzyme function".

Often asked as: explain how blood pH is controlled (5–6 marks); effect of exercise on blood pH (MCQ); why sodium citrate is added to a food or drink.

COMMON MISTAKE

- Saying a buffer keeps the pH **exactly** constant: during exercise blood pH falls slightly.
- Picking the wrong one of the two linked equilibria to apply.
- Saying food buffers are there to improve taste, colour or appearance.

⚠ Beyond the spec · not part of the Edexcel IAL specification

Examined in past papers even though the spec doesn't list it.

- The **HCO_3^- : H_2CO_3 ratio** needed to hold a given blood pH, calculated from Ka (WCH14 Jun 2021).

SPEC 14.23 Core Practical 11: finding Ka for a weak acid

WHAT THIS SPEC POINT ASKS

Determine Ka for a weak acid (Core Practical 11).

Principle

- At **half-neutralisation**, [HA] = [A^-], so **pH = pKa** (14.21).
- You make a half-neutralised solution by titrating one portion of the acid to the end point and then adding an **equal** portion of the same acid. You don't need to know the acid's concentration at all.

CORE PRACTICAL

Core practical 11: method in brief

1. **Calibrate the pH meter** with buffer solutions of known pH (e.g. pH 4.00 and 7.00), rinsing the electrode with distilled water between solutions.
2. **Pipette 25.0 cm³** of the weak acid (e.g. ethanoic acid, about 0.1 mol dm^{-3}) into a conical flask and add a few drops of **phenolphthalein**.
3. **Titrate** with about 0.1 mol dm^{-3} **NaOH** from a burette until the solution just turns **pale pink**. The acid is now exactly neutralised (all A^-).
4. Pipette **another 25.0 cm³** of the same acid into the flask and swirl. Now half the total acid is HA and half is A^-.
5. Measure the **pH** of the mixture with the pH meter. **pKa = pH**, and **Ka = 10^{-pH}**.
6. Repeat to get concordant values and take a mean.

Alternative: record pH every 1 cm³ (every 0.1 cm³ near the end point) while adding NaOH, using a pH meter or data logger; plot the titration curve and read the pH at half the equivalence volume.

Variables and data

Measured	Controlled	Calculated
pH of the half-neutralised mixture	temperature (Ka depends on it); equal volumes of acid in steps 2 and 4	Ka = 10^{-pH}, with units mol dm^{-3}

WORKED EXAMPLE

Worked example: processing the results

pH readings of the half-neutralised mixture: 4.78, 4.75, 4.76. Mean pH = 4.76.

pKa = 4.76; Ka = $10^{-4.76}$ = **1.7×10^{-5} mol dm^{-3}**.

A second method: a 0.100 mol dm^{-3} solution of the same acid had pH 2.88. Ka = $(10^{-2.88})^2 \div 0.100 = 1.74 \times 10^{-5}$ mol dm^{-3}. The two methods agree.

Safety

- **NaOH** is corrosive/irritant to eyes: wear eye protection. Ethanoic acid at this concentration is an irritant.
- Handle the glass pH electrode carefully; store it in its storage solution, not dry.

Sources of error and improvements

- **pH meter**: must be calibrated; reads to ±0.01 or ±0.02. A 0.02 error in pH is about a 5% error in Ka. Calibrate with a buffer close to the expected pH.
- **Temperature**: Ka (and the meter reading) changes with temperature; do the experiment at a constant, recorded temperature.
- **End point** judgement: overshooting with NaOH makes the mixture more than half-neutralised, so the pH (and pKa) reads high. Add NaOH dropwise near the end point.
- Rinse the electrode between readings and allow the reading to stabilise.

MARK-SCHEME WORDING

- "(when half the acid has been neutralised) [HA] = [A^-]" so "pH = pKa"; Ka = 10^{-pH}.
- Choice of indicator for the first titration: phenolphthalein, because its range lies in the vertical section of a weak acid–strong base curve.

Often asked as: explain why the pH at half-neutralisation equals pKa; read Ka from a recorded curve; select an indicator; evaluate the method (calibration, temperature).

COMMON MISTAKE

- Not recognising that the half-neutralisation pH gives pKa, even after doing the practical (examiners noted this).
- Using methyl orange for the first titration.
- Forgetting units for Ka (mol dm^{-3}) or giving it to 1 s.f.

Quick checks

QUICK CHECK

In $HSO_4^- + H_2O \rightleftharpoons SO_4^{2-} + H_3O^+$, which is the conjugate base of HSO_4^-?

- H_2O
- H_3O^+
- SO_4^{2-}
- H_2SO_4

HSO_4^- donates a proton and becomes SO_4^{2-}; the pair differs by one H^+. H_3O^+ is the conjugate acid of water.

QUICK CHECK

What is the pH of 0.0150 mol dm^{-3} potassium hydroxide at 298 K?

- 1.82
- 12.18
- 11.82
- 13.18

$[H^+] = 1.00 \times 10^{-14} \div 0.0150 = 6.67 \times 10^{-13}$ mol dm^{-3}, so pH = 12.18. 1.82 is the pOH, which is what you get if you take $-\log[OH^-]$ and stop.

QUICK CHECK

Pure water at 60 °C has pH 6.51. Which statement is correct?

- The water is slightly acidic
- K_w is smaller than at 25 °C
- $[H^+]$ is greater than $[OH^-]$
- The water is neutral and K_w is larger than at 25 °C

In pure water $[H^+]$ always equals $[OH^-]$, so it is neutral. Ionisation of water is endothermic, so K_w rises with temperature and the neutral pH falls below 7.

QUICK CHECK

What is the pH of 0.100 mol dm^{-3} of a weak acid with $K_a = 1.0 \times 10^{-5}$ mol dm^{-3}?

- 3.00
- 1.00
- 6.00
- 5.00

$[H^+] = \sqrt{(1.0 \times 10^{-5} \times 0.100)} = 1.0 \times 10^{-3}$ mol dm^{-3}, so pH = 3.00. 1.00 would be right only for a strong acid. 6.00 comes from forgetting the square root.

QUICK CHECK

Which indicator is suitable for titrating ammonia solution with hydrochloric acid?

- phenolphthalein, because the equivalence point is above 7
- methyl orange, because the steep section is roughly pH 3 to 7
- either, because both change colour near pH 7
- neither, because there is no steep section

A strong acid with a weak base has its steep section in the acidic region and an equivalence point near pH 5, which suits methyl orange. Phenolphthalein would change colour long before the equivalence point.

QUICK CHECK

25.0 cm^3 of a weak acid needs 24.0 cm^3 of NaOH for neutralisation. After 12.0 cm^3 of NaOH has been added the pH is 4.20. What is K_a?

- 4.20 mol dm^{-3}
- 1.6×10^{-2} mol dm^{-3}
- 6.3×10^{-5} mol dm^{-3}
- 4.0×10^{-9} mol dm^{-3}

12.0 cm^3 is half the neutralisation volume, so pH = pK_a = 4.20 and $K_a = 10^{-4.20} = 6.3 \times 10^{-5}$ mol dm^{-3}.

QUICK CHECK

Which mixture is a buffer?

- hydrochloric acid and sodium chloride
- sodium hydroxide and sodium chloride
- ethanoic acid and sodium hydroxide in a 1 : 1 mole ratio
- ammonia and ammonium chloride

A buffer needs a weak acid or weak base together with its conjugate partner. Ethanoic acid and NaOH in a 1 : 1 ratio react completely, leaving only sodium ethanoate; with excess ethanoic acid the mixture would be a buffer.

QUICK CHECK

Why is the enthalpy change of neutralisation of ethanoic acid with NaOH slightly less exothermic than that of HCl with NaOH?

- Ethanoic acid forms less water per mole
- Energy is used to dissociate the weak acid during the reaction
- Ethanoate ions react with water exothermically
- Ethanoic acid has two acidic hydrogens

Both reactions form one mole of water per mole of acid, but ethanoic acid is mostly undissociated, and breaking the O–H bond to release its H^+ takes in some energy.